

# goxpyriment — API Reference

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## goxpyriment API Reference

This guide documents the complete public API of the goxpyriment framework, organized by package.

Note that you can access the source code documentation by running pkgsite and opening <http://localhost:8080>. To install pkgsite:

```
go install golang.org/x/pkgsite/cmd/pkgsite@latest
```

See [Viewing the documentation locally](#) for details (and for serving this site offline with zensical).

### Package Overview

|            |                                                                  |
|------------|------------------------------------------------------------------|
| control/   | ← experiment lifecycle and orchestration (start here)            |
| stimuli/   | ← visual and audio stimulus objects                              |
| media/     | ← multi-movie playback with hardware-verified display onsets     |
| apparatus/ | ← SDL window/renderer, keyboard, mouse, gamepad, gamma corrector |
| results/   | ← experiment data file and output file                           |
| design/    | ← trial/block structure and randomization                        |
| clock/     | ← timing utilities                                               |
| geometry/  | ← coordinate conversion helpers                                  |

triggers/      ← hardware trigger devices (EEG sync, etc.)

---

## Package control

Import: `github.com/chrplr/goxpyriment/control`

## Boilerplate

Every experiment starts the same way:

```
exp := control.NewExperimentFromFlags("My Experiment", control.Black, control.White, 32)
defer exp.End()

err := exp.Run(func() error {
    // trial loop
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
```

## Pre-experiment Setup Dialog

`GetParticipantInfo` opens a graphical SDL window **before** the experiment starts to collect participant demographics, monitor properties, and display preferences. Call it before `NewExperiment` / `NewExperimentFromFlags`.

```
fields := append(control.StandardFields, control.FullscreenField)
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err) // user pressed Escape or closed window
}
```

```
// FieldType selects how a field is rendered.
type FieldType int
const (
    FieldText      FieldType = iota // text input box (default)
    FieldCheckbox                // tick-box; value is "true" or "false"
)

// InfoField describes one entry in the dialog.
type InfoField struct {
    Name      string // key in the returned map
    Label     string // displayed label
    Default   string // initial value
}
```

```

    Type      FieldType // FieldText (default) or FieldCheckbox
}

```

## Types

### Pre-built field sets

| Variable                  | Fields                                                |
|---------------------------|-------------------------------------------------------|
| control.ParticipantFields | subject_id, age, gender, handedness                   |
| control.MonitorFields     | screen_width_cm, viewing_distance_cm, refresh_rate_hz |
| control.StandardFields    | ParticipantFields + MonitorFields                     |
| control.FullscreenField   | Checkbox: fullscreen ("true" / "false")               |

### Function

| Function                                                                           | Description                                                                                                                  |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| GetParticipantInfo(title string, fields []InfoField)<br>(map[string]string, error) | Shows the dialog and returns collected values. Returns ErrCancelled if the user closes or presses Escape without confirming. |

**Session persistence** All values except subject\_id are saved to ~/.cache/goxpyriment/last\_session on OK and pre-filled on the next run. subject\_id is always reset.

```

info, err := control.GetParticipantInfo("My Experiment", fields)
// ...
fullscreen := info["fullscreen"] == "true"
width, height := 0, 0
if !fullscreen {
    width, height = 1024, 768
}
exp := control.NewExperiment("My Experiment", width, height, fullscreen,
    control.Black, control.White, 32)

// Set Info (and SubjectID) BEFORE Initialize – they are written to the info file autom
exp.SubjectID, _ = strconv.Atoi(info["subject_id"])
exp.Info = info

if err := exp.Initialize(); err != nil { log.Fatal(err) }
defer exp.End()

```

**Using the fullscreen checkbox and persisting to the data file** Initialize() writes a --PARTICIPANT INFO block to the companion -info.txt file whenever exp.Info is non-nil at that point. No explicit call to WriteParticipantInfo is needed.

```
control.ErrCancelled // returned when the user cancels the dialog
```

## Sentinel error

---

## Constructor Functions

| Function                                                                                                                | Description                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>NewExperimentFromFlags(name string, bg, fg Color, fontSize float32) *Experiment</code>                            | Creates and fully initializes an experiment from -w (windowed 1024×768), -d N (display index, -1 = primary), and -s N (subject ID) command-line flags. Calls <code>log.Fatal</code> on error. <b>This is the preferred entry point.</b> |
| <code>NewExperiment(name string, width, height int, fullscreen bool, bg, fg Color, fontSize float32) *Experiment</code> | Lower-level constructor; call <code>Initialize()</code> before use.                                                                                                                                                                     |

## Lifecycle Methods

| Method                                         | Description                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>exp.Initialize() error</code>            | Initializes SDL, audio, window, renderer, font, and data file.                                                                                                                                                                                                                                                                                       |
| <code>exp.End()</code>                         | Cleans up all resources. Always defer <code>exp.End()</code> immediately after construction.                                                                                                                                                                                                                                                         |
| <code>exp.Run(logic func() error) error</code> | Runs the main trial loop on the SDL main thread. Return <code>control.EndLoop</code> to exit cleanly.                                                                                                                                                                                                                                                |
| <code>exp.HideCursor() error</code>            | Hides the mouse cursor. <code>Initialize()</code> already does this — only needed to hide it again after a <code>ShowCursor</code> call.                                                                                                                                                                                                             |
| <code>exp.ShowCursor() error</code>            | Makes the mouse cursor visible. Call it after <code>Initialize()</code> in mouse-driven paradigms; the cursor is hidden by default so it never sits over the stimuli. Equivalent to setting <code>exp.CursorVisible = true</code> before <code>Initialize()</code> . <code>GetParticipantInfo</code> shows the cursor for its own dialog regardless. |

## Presentation Methods

| Method                                                                                                    | Description                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>exp.Show(stim VisualStimulus) error</code>                                                          | Clear → draw → flip. The standard one-call stimulus presentation.                                                                                                                                                                                                         |
| <code>exp.ShowTS(stim VisualStimulus) (uint64, error)</code>                                              | Clear → draw → flip, and return the SDL nanosecond timestamp captured immediately after the VSYNC flip. Use with <code>GetKeyEventTS</code> for hardware-precision RT measurement.                                                                                        |
| <code>exp.ShowTimed(stim VisualStimulus, durationMs int) error</code>                                     | <code>Show(stim) + Wait(durationMs)</code> in one call. For fixation crosses, cues, and passive stimulus viewing.                                                                                                                                                         |
| <code>exp.ShowFrames(stim VisualStimulus, n int) (uint64, error)</code>                                   | Hold the stimulus for exactly <i>n</i> display frames, returning the onset timestamp of the first flip. The stimulus is redrawn every frame — see <i>Holding a stimulus for a fixed number of frames</i> below.                                                           |
| <code>exp.BlankFrames(n int) (uint64, error)</code>                                                       | Frame-locked counterpart of <code>Blank(ms)</code> : clear and hold blank for exactly <i>n</i> frames, returning the timestamp of the first flip (i.e. the previous stimulus's offset).                                                                                   |
| <code>exp.ShowAndGetRT(stim VisualStimulus, keys []Keycode, timeoutMs int) (Keycode, int64, error)</code> | Clears stale keyboard events, shows stim with hardware-precise onset timing, waits for a key, and returns (key, rtMs, error). Pass <code>timeoutMs = -1</code> for no timeout; returns (0, 0, nil) on timeout. This is the canonical single-stimulus RT measurement call. |
| <code>exp.ShowEndMessage(message string) error</code>                                                     | Display a centered completion message and wait for any key. For end-of-experiment screens.                                                                                                                                                                                |
| <code>exp.ShowInstructions(text string) error</code>                                                      | Display centered text and wait for spacebar.                                                                                                                                                                                                                              |
| <code>exp.Blank(ms int) error</code>                                                                      | Clear and flip screen, then wait <i>ms</i> milliseconds.                                                                                                                                                                                                                  |
| <code>exp.Wait(ms int) error</code>                                                                       | Wait <i>ms</i> ms while pumping SDL events (ESC-abortable).                                                                                                                                                                                                               |
| <code>exp.ShowSplash(waitForKey bool) error</code>                                                        | Show experiment name + version splash.                                                                                                                                                                                                                                    |
| <code>exp.Flip() error</code>                                                                             | Present the backbuffer and hold to the frame boundary (one call = one display frame).                                                                                                                                                                                     |

## Input

| Method                                                 | Description                                                                                                          |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| exp.Keyboard                                           | *apparatus.Keyboard — see Keyboard section                                                                           |
| exp.Mouse                                              | *apparatus.Mouse — see Mouse section                                                                                 |
| exp.PollEvents(handle func(sdl.Event) bool) EventState | Process all pending SDL events; optionally forward to a handler. Returns EventState including nanosecond timestamps. |
| exp.HandleEvents() (Keycode, uint32, error)            | Convenience wrapper: returns (key, mouseButton, error).                                                              |

EventState now includes SDL event timestamps:

```
type EventState struct {
    LastKey          sdl.Keycode
    LastMouseButton  uint32
    LastKeyTimestamp uint64 // SDL nanosecond timestamp of the last key event
    LastMouseTimestamp uint64 // SDL nanosecond timestamp of the last mouse event
    QuitRequested    bool
}
```

## Design and Data

| Method                                                     | Description                                                                                                                     |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| exp.AddDataVariableNames(names []string)                   | Register CSV column names for the data file.                                                                                    |
| exp.Data.Add(values ...interface{})                        | Append a data row. Subject ID is prepended automatically.                                                                       |
| exp.AddBlock(b *design.Block, copies int)                  | Add trial blocks to the experiment.                                                                                             |
| exp.ShuffleBlocks()                                        | Randomize block presentation order.                                                                                             |
| exp.AddBWSFactor(name string, conditions []interface{})    | Register a between-subjects factor for Latin-square counterbalancing.                                                           |
| exp.GetPermutedBWSFactorCondition(name string) interface{} | Return this subject's condition for a BWS factor.                                                                               |
| exp.Design                                                 | *design.Experiment — full design object                                                                                         |
| exp.Info                                                   | map[string]string — values from GetParticipantInfo; set before Initialize() to persist them automatically to the -info.txt file |

| Method                         | Description                                                                                                                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>exp.CursorVisible</code> | <code>bool</code> — whether the mouse pointer is shown over the experiment window. Defaults to <code>false</code> : <code>Initialize()</code> hides the cursor. Set it to <code>true</code> before <code>Initialize()</code> (or call <code>ShowCursor()</code> after) for mouse-driven paradigms. |

## Font and Display

| Method                                                               | Description                                                         |
|----------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>exp.LoadFont(path string, size float32) error</code>           | Load a TTF font from file.                                          |
| <code>exp.LoadFontFromMemory(data []byte, size float32) error</code> | Load a TTF font from a byte slice.                                  |
| <code>exp.SetVSync(vsync int) error</code>                           | Toggle vertical sync (1 = on, 0 = off).                             |
| <code>exp.SetLogicalSize(w, h int32) error</code>                    | Set device-independent logical resolution.                          |
| <code>exp.SetOutputDirectory(dir string)</code>                      | Override default data file directory ( <code>~/goxpy_data</code> ). |

## Gamma Correction

Standard monitors apply a power-law transfer function  $L(V) = k \cdot (V/255)^\gamma$  ( $\gamma \approx 2.2$  for sRGB displays). Equal steps in RGB values do **not** produce equal steps in physical luminance. Use `SetGamma` to enable inverse-gamma correction.

| Method                                               | Description                                                                                   |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>exp.SetGamma(gamma float64)</code>             | Install a uniform inverse-gamma corrector. Call once after <code>Initialize()</code> .        |
| <code>exp.CorrectColor(c sdl.Color) sdl.Color</code> | Apply gamma correction to a color. Returns <code>c</code> unchanged when no corrector is set. |
| <code>exp.GammaCorrector</code>                      | <code>*apparatus.GammaCorrector</code> — set directly for per-channel calibration.            |

```
// Uniform gamma (typical sRGB monitor)
exp.SetGamma(2.2)

// Per-channel gamma (from photometer measurements)
exp.GammaCorrector = apparatus.NewGammaCorrector(2.1, 2.2, 2.3)

// Use in trial loop – specify colors in linear luminance space (0–255)
disk := stimuli.NewFilledCircle(exp.CorrectColor(control.RGB(128, 128, 128)), radius)
```

The `apparatus.GammaCorrector` type is also available directly:

```
gc := apparatus.NewGammaCorrectorUniform(2.2)
corrected := gc.CorrectColor(sdl.Color{R: 128, G: 128, B: 128, A: 255})
// corrected.R ≈ 186 – the physical digital value for 50% luminance on  $\gamma=2.2$ 
```

## Colors, Types, and Constants

```
// Named colors
control.Black, White, Red, Green, Blue, Yellow, Magenta, Cyan
control.Gray, DarkGray, LightGray

// Type aliases (so you only need to import "control")
type Color    = sdl.Color
type Keycode  = sdl.Keycode
type FPoint   = sdl.FPoint
type FRect    = sdl.FRect

// Constructors
control.RGB(r, g, b uint8) Color
control.RGBA(r, g, b, a uint8) Color
control.Point(x, y float32) FPoint
control.Origin() FPoint // returns (0, 0)

// Font helpers
control.FontFromFile(path string, size float32) (*ttf.Font, error)
control.FontFromMemory(data []byte, size float32) (*ttf.Font, error)

// Loop control
control.EndLoop           // sentinel error: return from Run callback to exit
control.IsEndLoop(err)    // test whether an error is the EndLoop sentinel

// Keyboard codes
control.K_SPACE, K_ESCAPE, K_RETURN, K_BACKSPACE, K_TAB
control.K_UP, K_DOWN, K_LEFT, K_RIGHT
control.K_A ... K_Z           // full alphabet
control.K_0 ... K_9           // digit row
control.K_KP_0 ... K_KP_9, K_KP_ENTER, K_KP_PLUS, K_KP_MINUS // numeric keypad
control.K_MINUS, K_PLUS, K_EQUALS, K_LEFTBRACKET, K_RIGHTBRACKET // punctuation

// Mouse buttons
control.BUTTON_LEFT, BUTTON_RIGHT
```

**Tip:** The full alphabet, digit row, keypad, and common punctuation keys are re-exported, so experiment code never needs to import `go-sdl3` just to name a key. For a rarely-used code not listed in `defaults.go`, import `go-sdl3/sdl` directly and use `sdl.K_...`



## Audio

```
exp.AudioDevice // sdl.AudioDeviceID – pass to Sound.PreloadDevice()

// Top-level helper (call before NewExperiment)
control.SetAudioSampleFrames(frames int) // set audio buffer size (256–2048)
```

## Microphone

```
exp.Microphone // *apparatus.Microphone – nil until OpenMicrophone is called

// Open the default recording device (nil = F32LE mono 44100 Hz)
exp.OpenMicrophone(spec *sdl.AudioSpec) error

// Closed automatically by exp.End()
```

---

## Package stimuli

Import: `github.com/chrplr/goxpyriment/stimuli`

## Interfaces

```
type Stimulus interface {
    Present(screen *apparatus.Screen, clear, update bool) error
    Preload() error
    Unload() error
}

type VisualStimulus interface {
    Stimulus
    Draw(screen *apparatus.Screen) error
    GetPosition() sdl.FPoint
    SetPosition(pos sdl.FPoint)
}
```

GPU textures are **lazily allocated** on the first Draw. To force early allocation (for timing-sensitive code), use:

```
stimuli.PreloadVisualOnScreen(screen, stim) // single stimulus
stimuli.PreloadAllVisual(screen, []VisualStimulus{...}) // slice
```

## Visual Stimuli

### Text

| Constructor                                                                         | Description                   |
|-------------------------------------------------------------------------------------|-------------------------------|
| <code>NewTextLine(text string, x, y float32, color Color) *TextLine</code>          | Single line of text.          |
| <code>NewTextBox(text string, width int32, pos FPoint, color Color) *TextBox</code> | Word-wrapped multi-line text. |

Both support a `Font *ttf.Font` field — set it to override the screen default.

## Shapes

| Constructor                                                              | Description                        |
|--------------------------------------------------------------------------|------------------------------------|
| <code>NewFixCross(size, lineWidth float32, color Color) *FixCross</code> | Fixation cross centered at (0, 0). |
| <code>NewCircle(radius float32, color Color) *Circle</code>              | Filled circle.                     |
| <code>NewRectangle(cx, cy, w, h float32, color Color) *Rectangle</code>  | Filled rectangle.                  |
| <code>NewLine(x1, y1, x2, y2 float32, color Color) *Line</code>          | Line segment.                      |

## Images and Video

| Constructor/Function                                                                                            | Description                                                                   |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <code>NewPicture(filePath string, x, y float32) *Picture</code>                                                 | Image loaded from file (PNG, JPG, BMP...).                                    |
| <code>NewPictureFromMemory(data []byte, x, y float32) *Picture</code>                                           | Image loaded from embedded bytes.                                             |
| <code>NewSpriteSheet(filePath string) *SpriteSheet</code>                                                       | One image holding many stimuli, sharing a single GPU texture.                 |
| <code>NewSpriteSheetFromMemory(data []byte) *SpriteSheet</code>                                                 | Same, from embedded bytes (required for the browser build).                   |
| <code>(*SpriteSheet) Grid(screen, cols, rows int) ([]*Sprite, error)</code>                                     | Cut into cols*rows equal cells, row-major.                                    |
| <code>(*SpriteSheet) GridWithSpacing(screen, cols, rows int, margin, spacing float32) ([]*Sprite, error)</code> | As Grid, for sheets with a border or gutters.                                 |
| <code>(*SpriteSheet) Sprites(screen, clips []sdl.FRect) ([]*Sprite, error)</code>                               | Explicit source rectangles, for irregular sheets.                             |
| <code>(*SpriteSheet) Unload() error</code>                                                                      | Destroys the shared texture. <code>Sprite.Unload</code> is a no-op by design. |
| <code>PlayGv(screen, path string, x, y float32) ([]UserEvent, error)</code>                                     | Play a .gv (LZ4-compressed RGBA) video file, VSYNC-locked.                    |

| Constructor/Function                      | Description                                |
|-------------------------------------------|--------------------------------------------|
| NewGvVideo(path string) (*GvVideo, error) | Open a .gv file for frame-by-frame access. |

## Psychophysics Stimuli

| Constructor                                                                                             | Description                                                                  |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| NewGaborPatch(sigma, theta, lambda, phase, psi, gamma float64, bgColor Color, size float32) *GaborPatch | Static Gabor patch. theta in degrees, lambda = spatial wavelength in pixels. |
| NewDotCloud(radius float32, bgColor, dotColor Color) *DotCloud                                          | Static random-dot cloud. Call Make(nDots, dotRadius, gap) to populate.       |
| NewRDS(imgSize, innerSize [2]int, shift, gap, scale int) *RDS                                           | Random-dot stereogram (side-by-side pair).                                   |
| NewVisualMask(w, h, dotW, dotH float32, bgColor, dotColor Color, pct int) *VisualMask                   | Random-dot masking stimulus. pct = dot fill percentage 0-100.                |

## Composite / Interactive

| Constructor                                                                                         | Description                                                                 |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| NewThermometerDisplay(size FPoint, nSegments int, state, goal float32) *ThermometerDisplay          | Segmented progress bar. State and Goal in 0-100.                            |
| NewChoiceGrid(choices []string, maxSelect int, prompt string) *ChoiceGrid                           | Multiple-choice button grid (mouse + keyboard). See below.                  |
| NewTextInput(msg string, pos FPoint, boxW float32, bgColor, frameColor, textColor Color) *TextInput | Free-text keyboard input box. Call ti.Get(screen, keyboard).                |
| NewMenu(items []string) *Menu                                                                       | Numbered keyboard-navigable list. Call m.Get(screen, keyboard, initialSel). |

## ChoiceGrid

```
cg := stimuli.NewChoiceGrid(choices, maxSelect, prompt)
cg.Cols = 7           // optional: set column count (0 = auto)

selections, err := cg.Get(exp.Screen, exp.Keyboard)
// selections is a []string preserving selection order
```

- MaxSelect > 0: auto-submits after N selections.
- MaxSelect == 0: participant presses ENTER or SPACE to submit.

- BACKSPACE removes the last selection.
- Both mouse click and matching keypress (single-char labels) activate buttons.

## Menu

```
m := stimuli.NewMenu([]string{"Option A", "Option B", "Option C"})
m.Pos = sdl.FPoint{X: 0, Y: 0} // optional: reposition (default = screen center)
m.HighlightColor = control.Yellow // optional: override highlight color

idx, err := m.Get(exp.Screen, exp.Keyboard, 0) // 0 = initially highlight first item
// idx is 0-based; -1 + sdl.EndLoop on ESC/quit
```

Navigation: UP/DOWN arrows move the highlight; ENTER or SPACE confirms; number keys 1–9 (0 for tenth) select and confirm directly. The selected item is shown in HighlightColor with a > prefix; others use TextColor. LineSpacing controls vertical item spacing (0 = auto from font height).

## Animated / VSYNC-locked Loops

All three functions disable GC, lock to VSYNC, and return (MotionResult, error).

```
type MotionResult struct {
    Key      sdl.Keycode // interrupt key pressed (0 if none)
    Button   uint8      // mouse button pressed (0 if none)
    RTms     int64      // ms from first frame to response (or total duration on timeout)
}
```

| Function                                                                                                                                                                                                                       | Description                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| PresentMovingDotCloud(screen, nDots int, dotRadius, cloudRadius float32, center FPoint, speedPxPerSec float32, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool, dotColor, bgColor Color) (MotionResult, error)   | Animated random-dot cloud. Each dot moves at a fixed speed and respawns when it exits the boundary. |
| PresentMovingGrating(screen, width, height float32, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error) | Drifting sinusoidal grating in a rectangular aperture.                                              |

| Function                                                                                                                                                                                                                           | Description                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| PresentMovingGabor(screen, size float32, sigma float64, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error) | Drifting Gabor patch with Gaussian envelope (alpha-blended edges). |

Spatial frequency is in **cycles per pixel** (e.g. 0.05 = one cycle every 20 px). Temporal frequency is in **Hz**. Orientation is in **degrees from horizontal** (0° = vertical bars drifting right).

### Stimulus Streams (High-Precision RSVP)

Stream functions disable GC, lock every onset and offset to a VSYNC boundary, and return ([]UserEvent, []TimingLog, error).

```

type UserEvent struct {
    Event      sdl.Event      // raw SDL event (KeyboardEvent, MouseButtonEvent, ...)
    Timestamp   time.Duration  // time relative to stream start (Go clock, ms precision)
    TimestampNS uint64         // SDL3 hardware event timestamp, nanoseconds (same clock)
}

type TimingLog struct {
    Index       int
    TargetOn     time.Duration
    ActualOnset  time.Duration // DIAGNOSTIC (Go clock): first-frame draw, stream-relat
    ActualOffset time.Duration // DIAGNOSTIC (Go clock): after last on-frame, stream-re
    OnsetNS      uint64         // AUTHORITATIVE: SDL3 ns timestamp (sdl.TicksNS clock)
    OffsetNS     uint64         // AUTHORITATIVE: SDL3 ns timestamp (sdl.TicksNS clock)
}

```

The OnsetNS/OffsetNS pair is the authoritative timing record: both are on the SDL3 nanosecond clock (sdl.TicksNS()) — the same clock that stamps input events (UserEvent.TimestampNS, Keyboard.GetKeyEventTS) and VSYNC flips (Screen.FlipTS). A reaction time or a displayed duration is a plain subtraction within this one clock: int64(event.TimestampNS - l.OnsetNS), or int64(l.OffsetNS - l.OnsetNS).

ActualOnset/ActualOffset are Go-clock (time.Now) **diagnostics only**, kept for coarse pacing checks. They live on a different timebase (different origin) from the NS fields and from input events, so they must never be subtracted from an event timestamp or logged as a canonical onset/offset. Use the NS fields for any real timing.

OnsetNS is zero only when a stimulus was never displayed (zero on-frames); OffsetNS is the VSYNC timestamp of the flip that takes the stimulus down, and for the final element of a contiguous stream (which has no ISI flip of its own) it is synthesised from

the onset plus the on-frame count so that it, too, is on the SDL clock rather than one frame early.

UserEvent.TimestampNS and TimingLog.OnsetNS/OffsetNS are all on the SDL3 nanosecond clock, so reaction times measured during a stream can be computed with full hardware precision:

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        // Find the stimulus that was on-screen when the key was pressed
        for _, l := range logs {
            if ev.TimestampNS >= l.OnsetNS && ev.TimestampNS < l.OffsetNS {
                rtNS := int64(ev.TimestampNS - l.OnsetNS)
                fmt.Printf("RT from stimulus %d: %d ms\n", l.Index, rtNS/1_000_000)
            }
        }
    }
}
```

## Searching event lists

| Function                                                             | Description                                                                      |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------|
| FirstKeyPress(events []UserEvent, key sdl.Keycode) (UserEvent, bool) | Returns the first KEY_DOWN event matching key from the slice, plus a found flag. |

```
if ev, ok := stimuli.FirstKeyPress(events, sdl.K_SPACE); ok {
    fmt.Printf("Space pressed at %d ms\n", ev.Timestamp.Milliseconds())
}
```

```
// RSVP text stream – simplest entry point
events, logs, err := stimuli.PresentStreamOfText(
    exp.Screen, words, durationOn, durationOff, x, y, color,
)

// Image/mixed stream
elements := stimuli.MakeRegularVisualStream(stims, durationOn, durationOff)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x, y)

// Irregular timing
elements, err := stimuli.MakeVisualStream(stims, onsetMs, durationMs)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x, y)
```

## Visual Streams

```
// Regular timing
elements := stimuli.MakeRegularSoundStream(sounds, durationOn, durationOff)
events, logs, err := stimuli.PlayStreamOfSounds(elements)

// Irregular timing
elements, err := stimuli.MakeSoundStream(sounds, onsetMs, durationMs)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

**Audio Streams** `sounds` is `[]stimuli.AudioPlayable` — satisfied by both `*Sound` and `*Tone`. The returned `TimingLog` carries `OnsetNS/OffsetNS` on the SDL3 clock (captured at the `Play()` instant and the end of the on-phase), so audio-stream onsets are directly comparable with input-event timestamps, same as visual streams.

**Mixed Streams** `PresentStreamOfStimuli` presents a heterogeneous, **sequential** stream that freely mixes visual and audio stimulus types (anything satisfying the broad `Stimulus` interface). It uses the same VSYNC-locked, GC-disabled loop as `PresentStreamOfImages` (which now delegates to it).

```
elements := []stimuli.StreamElement{
    {Stimulus: stimuli.NewTextLine("Ready?", 0, 0, color), DurationOn: 600 * time.Millisecond},
    {Stimulus: pic, DurationOn: 800 * time.Millisecond}, // held while the tone plays
    {Stimulus: tone, DurationOn: 400 * time.Millisecond, DurationOff: 200 * time.Millisecond},
}
events, logs, err := stimuli.PresentStreamOfStimuli(exp.Screen, elements, x, y)

// Builders mirror the visual/sound ones but take []stimuli.Stimulus:
elements = stimuli.MakeRegularStream(stims, durationOn, durationOff)
elements, err = stimuli.MakeStream(stims, onsetMs, durationMs)
```

Per-element semantics:

- **Visual** elements (those satisfying `VisualStimulus`) are centered on `(x, y)` and redrawn every frame for `DurationOn`, then blanked for `DurationOff`.
- **Audio / non-visual** elements (and a `nil Stimulus`) are triggered once, right after the slot's first VSYNC flip, and **do not clear the screen** — the previous frame is held for the whole slot. Place a visual just before a sound to keep it on screen while the sound plays. `TimingLog.OnsetNS` is the VSYNC reference at which the audio was triggered.

Audio elements must already be bound to the device via `PreloadDevice(exp.AudioDevice)` before the call (same precondition as `PlayStreamOfSounds`). For pure-audio streams prefer `PlayStreamOfSounds`, whose sub-frame sleep-polling gives finer timing than 60 Hz frame quantization.

**Per-frame callback** `PresentStreamOfStimuliFunc` adds an optional `FrameCallback` invoked once per frame, **just before each flip**. Use it to draw a persistent overlay (trial counter, frame border, fixation) on top of the stimulus, or to run real-time logic

such as firing feedback the instant a response window lapses — things the plain stream functions cannot express.

```
header := stimuli.NewTextLine("3 / 40", 0, -300, control.White)
cb := func(ctx stimuli.FrameContext) error {
    _ = header.Draw(ctx.Screen)           // overlay, rendered over the stimulus
    if ctx.OnPhase && ctx.FirstFrame {    // onset of element ctx.Index
        // real-time feedback using ctx.NowNS / ctx.Events;
        // return sdl.EndLoop to stop early, any other error to abort
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliFunc(exp.Screen, elements, x, y, cb)
```

FrameContext fields: Screen, Index, Frame, OnPhase, FirstFrame, NowNS (pre-flip `sdl.TicksNS()`), Elapsed, Events (through the previous frame). Passing nil for the callback is identical to `PresentStreamOfStimuli`.

**Per-stimulus onset trigger (post-flip hook)** To emit a hardware TTL — or run any action — aligned to each stimulus **onset**, use `PresentStreamOfStimuliHooks`, which adds an optional `OnsetCallback` to the per-frame callback:

```
func PresentStreamOfStimuliHooks(
    screen *apparatus.Screen, elements []StreamElement, x, y float32,
    onFrame FrameCallback, onOnset OnsetCallback,
) ([]UserEvent, []TimingLog, error)

type OnsetCallback func(index int, onsetNS uint64) error
```

`onOnset` fires once per element with a stimulus onset, **immediately after** the VSYNC flip that turns it on (for a held audio element whose `DurationOn` rounds to zero frames, after the first-off-frame flip that triggers it). `index` matches the returned `TimingLog` slice and `onsetNS` equals `TimingLog[index].OnsetNS`.

This is the **post-flip** counterpart to `FrameCallback`: `FrameCallback` runs one frame earlier (pre-flip, at GPU-submission time), whereas `OnsetCallback` runs on the photon side of the frame boundary, so a trigger emitted here coincides with the logged onset instead of leading it by a frame.

```
dev := triggers.NewParallelPort("/dev/parport0") // any triggers.OutputTTLDevice
if err := dev.Open(); err != nil {
    log.Fatal(err)
}
defer dev.Close()

onset := func(index int, onsetNS uint64) error {
    return dev.Send(codes[index]) // per-stimulus code, at the displayed onset
}

clear := func(ctx stimuli.FrameContext) error {
    if !ctx.OnPhase && ctx.FirstFrame { // drop the lines at the start of each ISI

```



```

        return dev.AllLow()
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliHooks(exp.Screen, elements, 0, 0, cl

```

The hook runs on the timing-critical path with GC disabled, between the onset flip and the next frame, so it **must be short and non-blocking**: emit an edge (SetHigh / Send) here and clear it from a later FrameCallback or after the stream — never time.Sleep or a blocking Pulse, or a frame is missed. Elements with a nil Stimulus (pure delay / hold slots) have no onset and do not fire it. Either callback may be nil; PresentStreamOfStimuliFunc is exactly PresentStreamOfStimuliHooks(..., onFrame, nil).

For pure-audio streams the equivalent is PlayStreamOfSoundsHook(elements, onOnset), which fires onOnset once per non-nil Sound, immediately after Play(), with the same SDL-clock onsetNS (== TimingLog[index].OnsetNS). PlayStreamOfSounds is PlayStreamOfSoundsHook(elements, nil).

## Audio Stimuli

```

// WAV file
snd := stimuli.NewSound(filePath)
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
snd.Wait() // block until done
snd.PlaySegment(onset, offset, rampSec) // time-delimited segment with optional f

// Embedded WAV
snd := stimuli.NewSoundFromMemory(data)

// Procedural tone
tone := stimuli.NewTone(frequency, duration, volume) // duration: time.Duration; volu
tone.PreloadDevice(exp.AudioDevice)
tone.Play()

// One-shot helper (no preload needed)
stimuli.PlaySoundFromMemory(exp.AudioDevice, data)

// Embedded feedback sounds (via assets_embed)
import "github.com/chrplr/goxpyriment/assets_embed"
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.BuzzerWav)
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.CorrectWav)

```

## Package apparatus and results

Import: `github.com/chrplr/goxpyriment/apparatus` (screen, input, gamma) Import: `github.com/chrplr/goxpyriment/results` (data file)

In normal experiments you access apparatus types through `exp.Screen`, `exp.Keyboard`, `exp.Mouse`, and `exp.Data`. Direct use of apparatus is only needed when writing custom stimulus types.

### Screen

All stimulus positions use a **center-origin coordinate system**: (0, 0) is the screen center; positive Y is upward.

```
screen.CenterToSDL(x, y float32) (float32, float32) // convert to SDL top-left coords
screen.CenteredRect(pos FPoint, w, h float32) *FRect // SDL dest rect for a w×h texture
screen.MousePosition() (float32, float32) // current cursor in center coord
screen.Clear() error // fill with background color
screen.Update() error // present + hold to the frame boundary
screen.Flip() error // alias for Update
screen.FlipTS() (uint64, error) // present + return SDL nanosecond timestamp
screen.FrameDuration() time.Duration // nominal frame duration (falls on frame boundary)
screen.CalibrateRefresh(n int) (time.Duration, error) // measured frame period, pacing
screen.SetLogicalSize(w, h int32) error
screen.SetVSync(vsync int) error
screen.DisplayInfo() apparatus.DisplayInfo // monitor properties
screen.Destroy()
```

`FlipTS` returns `sdl.TicksNS()` captured immediately after the flip. This timestamp is on the same nanosecond clock as SDL3 event timestamps, so `int64(event.Timestamp - onsetNS)` gives hardware-precision reaction time without any polling latency.

`Update` (and therefore `Flip`, `FlipTS`, `Show`, `ShowTS`) presents **and then holds to the frame boundary**, so one call always occupies exactly one display frame. This is unconditional because `SDL_RenderPresent` cannot be trusted to block until the retrace — under triple/mailbox buffering it returns immediately — and the wait costs nothing where the driver does block. Pacing is skipped when `VSync` is off.

`CalibrateRefresh` bypasses that pacing and presents directly, so it reports the *unaided* driver behaviour; compare it against `FrameDuration`. `Initialize` runs it over 60 frames at startup and writes `sys_refresh_nominal_hz / sys_refresh_measured_hz` into the session's `-info.txt`, warning if they differ by more than 10%. Measured slower than nominal means frames are being dropped before the panel, which pacing cannot fix.

**Holding a stimulus for a fixed number of frames** There is no “wait for *n* VSYNC edges” call, and there cannot be a useful one: SDL3 exposes no `SDL_WaitVBlank`, so the only way to stay locked to the display is to present a frame — and a frame carrying no draw calls is not reliably scanned out under a compositor (see *Never present a*

*frame with no draw calls* in apparatus/CLAUDE.md). A hold must therefore **redraw its content once per frame**:

```
for i := 0; i < n; i++ {
    screen.Clear()
    stim.Draw(screen)
    screen.Flip()
}
```

`exp.ShowFrames(stim, n)` and `exp.BlankFrames(n)` wrap that loop and return the on-set timestamp of the first flip:

```
onsetNS, _ := exp.ShowFrames(rect, 10) // 10 frames on screen
offsetNS, _ := exp.BlankFrames(10)     // 10 frames blank
exp.Data.Add(onsetNS, offsetNS)
```

There is no “wait for n frames” call that skips the redraw. A hold must redraw its content once per frame: SDL invalidates the backbuffer on every present, so re-clearing with whatever draw colour was last set paints the screen in that colour instead of holding the stimulus. Use the two calls above.

## Keyboard

```
key, err := exp.Keyboard.Wait() // any key
key, err := exp.Keyboard.WaitKey(control.K_SPACE) // specific key
key, err := exp.Keyboard.WaitKeys(keys, timeoutMS) // first of several keys
key, rt, err := exp.Keyboard.WaitKeysRT(keys, timeoutMS) // with RT in ms from onset
key, ts, err := exp.Keyboard.GetKeyEventTS(keys, timeoutMS) // with SDL event timestamp
events, err := exp.Keyboard.GetKeyEventsTS(keys, timeoutMS) // first key + 50 ms
events, err := exp.Keyboard.CollectKeyEventsTS(keys, durationMS) // all keys during full hold
key, err := exp.Keyboard.Check() // non-blocking poll
held := exp.Keyboard.IsPressed(key) // true if key is physically pressed
upTS, err := exp.Keyboard.WaitKeyReleaseTS(key, timeoutMS) // wait for KEY_UP; r
exp.Keyboard.Clear() // drain SDL event queue
```

`WaitKeys` and `WaitKeysRT` return 0, nil on timeout; return `sdl.EndLoop` on ESC or window close.

`IsPressed` queries SDL’s scancode state array — no event queue involvement. `WaitKeyReleaseTS` returns the `KEY_UP` hardware timestamp so that `upTS` - `downTS` gives nanosecond-precision press duration.

**GetKeyEventsTS — two-phase collection for bilateral responses.** The timeout governs how long to wait for the *first* key. After the first key arrives, the function waits an additional 50 ms for any second key before returning. This extra window is necessary because human “simultaneous” bilateral presses (e.g. both hands at once) arrive 10–50 ms apart — a non-blocking drain after the first key would miss the second key almost every time. Use `GetKeyEventTS` for ordinary single-key trials to avoid the 50 ms overhead.

**GetKeyEventTS** returns the `SDL3 KeyboardEvent.Timestamp` field — the nanosecond

time at which the hardware key-down event was generated, on the same clock as `sdl.TicksNS()` and `Screen.FlipTS()`. This allows computing reaction time from any specific stimulus onset without manual arithmetic:

```
onset, _ := exp.ShowTS(stim1)    // nanoseconds at VSYNC flip
exp.Wait(500)
exp.ShowTS(stim2)
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
rtToStim1 := int64(eventTS - onset) // nanoseconds
```

## Mouse

```
x, y := exp.Mouse.Position()           // current position (center of screen)
btn, err := exp.Mouse.WaitPress()       // block until button pressed
btn, rt, err := exp.Mouse.WaitPressRT(timeoutMS) // with RT in ms from call site
btn, ts, err := exp.Mouse.GetPressEventTS(timeoutMS) // with SDL event timestamp (nanoseconds)
btn, err := exp.Mouse.Check()           // non-blocking poll
held := exp.Mouse.IsPressed(sdl.BUTTON_LEFT) // true if button is physically pressed
upTS, err := exp.Mouse.WaitButtonReleaseTS(btn, timeoutMS) // wait for MOUSE_BUTTON_UP;
exp.Mouse.ShowCursor(show bool) error
```

`WaitPressRT` mirrors `Keyboard.WaitKeysRT`: reaction time is measured in milliseconds from the call site. `GetPressEventTS` returns the SDL3 hardware event timestamp in nanoseconds, suitable for use with `ShowTS`. `IsPressed` and `WaitButtonReleaseTS` mirror the keyboard's `IsPressed` and `WaitKeyReleaseTS`.

## GamePad

```
pads, err := apparatus.GetGamePads() // enumerate connected gamepads
defer pads[0].Close()

btn, err := pads[0].WaitPress() // block until any button pressed
btn, ts, err := pads[0].GetPressEventTS(timeoutMS) // with SDL event timestamp
```

`GetPressEventTS` returns the `GamepadButtonEvent.Timestamp` field — same nanosecond clock as `Screen.FlipTS` and keyboard/mouse event timestamps.

## Unified Input — `WaitAnyEventTS`

When the response device is not fixed in advance (keyboard or mouse click), use the method on `Experiment`:

```
// Accept F or J key, or any mouse button, timeout after 3 s
ev, err := exp.WaitAnyEventTS(
    []control.Keycode{control.K_F, control.K_J},
    true, // catchMouse
    3000,
)
```

Returns an `apparatus.InputEvent`:

```
type InputEvent struct {
    Device      apparatus.DeviceKind // DeviceKeyboard | DeviceMouse | DeviceGame
    Key         sdl.Keycode // non-zero for keyboard events
    Button      uint32 // non-zero for mouse events
    GamepadButton sdl.GamepadButton // non-zero for gamepad events
    TimestampNS uint64 // SDL3 hardware timestamp, nanoseconds
}
```

`TimestampNS` is on the same clock as `ShowTS`, so RT computation is identical regardless of device:

```
onset, _ := exp.ShowTS(stim)
ev, _ := exp.WaitAnyEventTS(keys, true, -1)
rtNS := int64(ev.TimestampNS - onset)
```

Pass `keys = nil` to accept any key. Pass `catchMouse = false` to ignore the mouse. On timeout, returns a zero `InputEvent` and `nil` error. On ESC or quit, returns `sdl.EndLoop`.

## ResponseDevice

`ResponseDevice` is a unified interface over all participant-input hardware — SDL-event-driven devices (keyboard, mouse, gamepad) **and** polled TTL devices (MEGTTLBox, DLPIO8). It is the recommended abstraction when the experiment design does not commit to a specific input modality.

```
type ResponseDevice interface {
    WaitResponse(ctx context.Context) (Response, error)
    DrainResponses(ctx context.Context) error
    Close() error
}

type Response struct {
    Source apparatus.DeviceKind // DeviceKeyboard | DeviceMouse | DeviceGamepad | Dev
    Code   uint32 // SDL Keycode, mouse button, gamepad button, or TTL bitmask
    RT     time.Duration // elapsed from WaitResponse call to detection
    Precise bool // true = hardware event timestamp; false = software poll
}
```

`Response.Precise` distinguishes two timing regimes:

| Device                   | Precise | RT origin                                                                     |
|--------------------------|---------|-------------------------------------------------------------------------------|
| Keyboard, Mouse, Gamepad | true    | SDL3 hardware event timestamp (nanosecond)                                    |
| MEGTTLBox, DLPIO8        | false   | <code>time.Now()</code> at poll detection ( $\pm$ poll interval, $\sim 5$ ms) |

Construct wrappers with the provided adapters:

```
// SDL-event-driven devices
rd := &apparatus.KeyboardResponseDevice{KB: exp.Keyboard}
rd := &apparatus.MouseResponseDevice{M: exp.Mouse}
rd := &apparatus.GamepadResponseDevice{GP: pad}

// Polled TTL device (MEGTTLBox, DLPI08, or any type with ReadAll/DrainInputs)
box, _ := triggers.NewMEGTTLBox("/dev/ttyACM0")
rd := apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
```

Usage in a trial loop:

```
onset, _ := exp.ShowTS(stim)
_ = rd.DrainResponses(ctx)
resp, err := rd.WaitResponse(ctx)
// resp.RT is always valid; resp.Precise tells you whether to trust nanosecond accuracy
```

## Microphone

```
// Construction
mic, err := apparatus.NewMicrophone(spec) // default recording
mic, err := apparatus.NewMicrophoneFromDevice(devID, spec) // specific device
apparatus.DefaultRecordingSpec() sdl.AudioSpec // F32LE mono 44100
devices, err := apparatus.GetRecordingDevices() []sdl.AudioDeviceID

// Fields
mic.DeviceID sdl.AudioDeviceID
mic.Stream *sdl.AudioStream
mic.Spec sdl.AudioSpec

// Methods
captureStartNS, err := mic.StartCapture() // flush buffer, resume device, return Tic
err := mic.StopCapture() // pause device
n, err := mic.ReadSamples(buf []byte) // read PCM bytes (non-blocking)
n, err := mic.Available() // bytes waiting to be read
mic.Close() // destroy stream
```

In normal experiments, open the microphone via `exp.OpenMicrophone(spec)` (see control package); `exp.Microphone` is then available throughout the experiment and closed by `exp.End()`.

## VoiceKey

Detects voice onset by computing per-window RMS over F32LE PCM samples.

```
vk := apparatus.NewVoiceKey(mic, threshold float32, windowSz int) *VoiceKey
// threshold: RMS amplitude (0–1). Recommended starting value: 0.02–0.05.
// windowSz: samples per RMS window. 0 → 128 (≈ 2.9 ms at 44100 Hz).
```

```
captureStartNS, err := vk.Arm() // flush mic buffer, start capture, return TicksNS()

onsetNS, pcm, err := vk.WaitOnset(captureStartNS uint64, timeoutMS int) (uint64, []byte)
// Returns: onset timestamp (sdl.TicksNS() domain), all PCM bytes captured, nil or ErrV
```

onsetNS and the imageOnsetNS returned by exp.ShowTS or screen.FlipTS are on the same SDL3 nanosecond clock:

```
captureStartNS, _ := vk.Arm()
imageOnsetNS, _ := exp.ShowTS(picture)
onsetNS, pcm, _ := vk.WaitOnset(captureStartNS, 2000)
rtMs := int64(onsetNS - imageOnsetNS) / 1_000_000
```

Sentinel error: apparatus.ErrVoiceTimeout

Post-hoc helpers (for re-analysing saved WAV files):

```
sampleIdx := apparatus.ScanOnset(pcm []byte, threshold float32, windowSz int) int
// Returns the sample index of the first onset window, or -1 if not found.

onsetNS := apparatus.SampleOnsetNS(captureStartNS uint64, sampleIndex, sampleRate int)
// Converts a sample index back to a nanosecond timestamp.

names, err := apparatus.DeviceNames() map[sdl.AudioDeviceID]string
// Returns all recording devices and their human-readable names.
```

## WriteWAV

```
apparatus.WriteWAV(path string, spec sdl.AudioSpec, data []byte) error
```

Saves raw PCM bytes as a standard RIFF/WAV file. Supports AUDIO\_F32LE, AUDIO\_S16LE, AUDIO\_S32LE, and AUDIO\_U8. Writes atomically via a temp file + rename.

---

## DataFile

```
exp.Data.Add(field1, field2, ...) // append a data row
exp.Data.AddVariableNames([]string{...}) // write column header
exp.Data.WriteDisplayInfo(info) // append display metadata to the info fi
exp.Data.WriteHostInfo(sysinfo.Host()) // append machine/OS metadata to the info
exp.Data.WriteParticipantInfo(info) // append --PARTICIPANT INFO block to the
exp.Data.WriteEndTime() // append end time + duration to the info
```

Two files are written to ~/goxpy\_data/ for each session:



| File                                                  | Contents                                                                                                                |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv      | Pure CSV data rows — directly importable by Excel, R, or pandas                                                         |
| <expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt | #-prefixed metadata: start/end time, hostname, OS, framework version, display and audio configuration, participant info |

## Package media

Import: `github.com/chrplr/goxpyriment/media`

Multi-movie playback with shared master-clock synchronisation, look-ahead frame conditions, and post-vsync display events for hardware- trigger alignment. Adds MovieManager (per-experiment owner of the clock and movie set), Movie (per-movie state with PsyScope-Tahoe semantics), and the media/present subpackage of OS-level vsync backends. Pure Go, .gv movies only, silent.

**Full reference (precision contracts, platform support, mapping to PsyScope movie commands):** see [MediaMovies.md](#).

## Quick API summary

```
mgr := media.NewMovieManager(exp.Screen)
defer mgr.Close()

gv, _ := gvvideo.LoadGVVideo("clip.gv")
mov, _ := media.NewMovie(mgr, gv,
    media.WithTag("M1"),
    media.WithRepeat(-1),
    media.WithSize(640, 360),
    media.WithPosition(sdl.FPoint{X: -200, Y: 0}),
)
defer mov.Close()

// Look-ahead – fires from inside DrawWithoutFlip, BEFORE the frame is presented.
mov.OnAt(media.Frame(186), func(o media.Onset) { /* ... */ })

// Hardware-verified – fires from NotifyFlipped with the OS vsync timestamp.
mov.OnAtDisplay(media.Frame(186), func(o media.Onset) {
    _ = ttl.Pulse(0, 5*time.Millisecond)
})

// Display[Onset/Offset] – fires when the named tag changes visibility.
mgr.OnDisplayOnset("M1", func(o media.Onset) { /* ... */ })
```



```
// Atomic command burst (PsyScope script-line equivalent):
mgr.BeginBurst()
movieA.Pause(); movieB.Pause(); movieC.Pause()
mgr.EndBurst()

// Per-frame, mixed compositing:
exp.Screen.Clear()
mgr.DrawWithoutFlip()           // decode + render movies
fix.Present(exp.Screen, false, false) // overlay
ts, _ := exp.Screen.FlipTS()
mgr.NotifyFlipped(ts)
```

## Onset precision summary

| Source           | Where                                             | Meaning                                 |
|------------------|---------------------------------------------------|-----------------------------------------|
| LookAhead        | OnAt / OnDone callbacks                           | sdl.TicksNS() at fire time, pre-vsync   |
| VsyncEstimated   | OnAtDisplay / Display* callbacks (fallback timer) | Post-Present FlipTS, ~vsync-period      |
| HardwareVerified | OnAtDisplay / Display* callbacks (Stage 5 timer)  | OS-measured first-pixel-visible, sub-ms |

## Platform status

| Platform | Backend                                             | Precision        |
|----------|-----------------------------------------------------|------------------|
| macOS    | CVDisplayLink (CoreVideo via purego)                | HardwareVerified |
| Linux    | DRM_IOCTL_WAIT_VBLANK (/dev/dri/cardN, video group) | HardwareVerified |
| Windows  | (not yet — fallback)                                | VsyncEstimated   |
| Other    | (fallback)                                          | VsyncEstimated   |

See [MediaMovies.md §8](#) for the Windows roadmap and [§7](#) for the trigger-synchrony contract (the wire pulse arrives ~0.5–5 ms after Onset.TimestampNS; calibrate the constant offset with a photodiode for sub-ms wire-side alignment).

## Package design

Import: [github.com/chrplr/goxpyriment/design](https://github.com/chrplr/goxpyriment/design)

## Data Structures

```
// Trial – one experimental trial
trial := design.NewTrial()
trial.SetFactor("condition", "congruent")
trial.GetFactor("condition") // → "congruent"
trial.Copy()                // deep copy

// Block – a sequence of trials
block := design.NewBlock("Practice")
block.SetFactor("type", "practice")
block.AddTrial(trial, copies, randomPosition)
block.ShuffleTrials()

// Experiment design (separate from control.Experiment)
exp.Design // *design.Experiment – contains Blocks, DataVariableNames, etc.
```

## Randomization

```
design.RandInt(a, b int) int           // random int in [a, b]
design.RandElement(list []T) T         // random element (generic)
design.CoinFlip(headBias float64) bool // weighted coin flip
design.RandNorm(a, b float64) float64  // truncated normal in [a, b]
design.ShuffleList(list []T)           // in-place Fisher-Yates shuffle (g
design.MakeMultipliedShuffledList(list []T, n int) []T // n shuffled copies concatenated
design.RandIntSequence(first, last int) []int // shuffled range [first, last]
```

## Latin Square (Between-Subjects Counterbalancing)

```
// Registration
exp.AddBWSFactor("handedness", []interface{}{"left", "right"})

// At runtime – returns the condition for the current subject
condition := exp.GetPermutedBWSFactorCondition("handedness")

// Low-level
square, err := design.LatinSquare(elements, design.PBalancedLatinSquare)
square, err := design.LatinSquareInts(n, design.PCycledLatinSquare)
// permutation types: design.PBalancedLatinSquare, PCycledLatinSquare, PRandom
```

---

## Package clock

Import: [github.com/chrplr/goxpyriment/clock](https://github.com/chrplr/goxpyriment/clock)

```

clock.Wait(ms int)           // block for ms milliseconds
clock.GetTime() int64        // ms since package first used
clock.GetTimeNS() int64      // nanoseconds since package first used

c := clock.NewClock()        // clock relative to "now"
c.NowMillis() int64          // ms elapsed
c.NowNanos() int64           // nanoseconds elapsed
c.Now() time.Duration
c.Reset()                   // restart reference
c.SleepUntil(target time.Duration) // sleep until target offset (returns immediately)

```

**Note:** Prefer `exp.Wait(ms)` over `clock.Wait(ms)` in experiment code — `exp.Wait` pumps SDL events and detects ESC.

**Clock domains:** `GetTimeNS()` and `NowNanos()` use the Go monotonic clock (`time.Since`). SDL event timestamps from `Screen.FlipTS`, `GetKeyEventTS`, `GetPressEventTS`, and `WaitAnyEventTS` use `sdl.TicksNS()`. The two clocks have different origins and **must not be subtracted from each other** for reaction-time computation. Use the SDL-based functions exclusively for RT measurement.

---

## Package geometry

Import: `github.com/chrplr/goxpyriment/geometry`

```

geometry.GetDistance(p1, p2 sdl.FPoint) float32
geometry.CartesianToPolar(x, y float32) (radius, angleDeg float32)
geometry.PolarToCartesian(radius, angleDeg float32) (x, y float32)
geometry.DegreeToRadian(deg float32) float64

```

---

## Package triggers

Import: `github.com/chrplr/goxpyriment/triggers`

Provides hardware TTL signal output (EEG/MEG trigger codes) and TTL input (response pads wired over serial). Lines are **0-indexed (0-7)** throughout; bit N of a bitmask corresponds to line N.

## Interfaces

```

// OutputTTLDevice – send trigger codes to recording equipment.
type OutputTTLDevice interface {
    Send(mask byte) error           // set all 8 lines from bitmask
    SetHigh(line int) error         // drive line HIGH (0-indexed)
    SetLow(line int) error          // drive line LOW  (0-indexed)
}

```

```

    Pulse(line int, d time.Duration) error // HIGH for d, then LOW (blocks)
    AllLow() error                          // all lines LOW
    Close() error                           // AllLow + release port
}

// InputTTLDevice – read TTL inputs from response hardware.
type InputTTLDevice interface {
    ReadAll() (byte, error) // bitmask of
    ReadLine(line int) (byte, error) // 0 or 1 (0-i
    WaitForInput(ctx context.Context) (mask byte, rt time.Duration, err error)
    DrainInputs(ctx context.Context) error
    Close() error
}

```

## DLPIO8 (DLP-IO8-G, USB-CDC serial)

Implements both OutputTTLDevice and InputTTLDevice.

```

// Auto-detect (recommended): returns NullOutputTTLDevice if not found, no error.
out, portName, err := triggers.AutoDetectDLPIO8()
defer out.Close()
out.Pulse(0, 10*time.Millisecond) // 10 ms pulse on line 0

// Manual:
d, err := triggers.NewDLPIO8("/dev/ttyUSB0")
defer d.Close()
d.Send(0b00000101) // lines 0 and 2 HIGH
mask, err := d.ReadAll() // bitmask of all 8 input lines
mask, rt, err := d.WaitForInput(ctx)

```

## DLPIO20 (DLP-IO20, USB-CDC serial)

Implements both OutputTTLDevice and InputTTLDevice. 5 V logic (the T4 is 3.3 V).

**Untested on hardware** — written from the [datasheet](#) rev 1.1. Bring it up with tests/test\_dlpio20 and a multimeter before relying on it.

The device has 17 usable digital channels but the TTL interfaces are 8-bit, so interface lines 0–7 address a *window* of 8 channels — AN0–AN7 for output and AN8–AN13, RB6, RB7 for input by default. Channels outside the windows stay reachable through the channel-level methods.

```

// Auto-detect (recommended): returns NullOutputTTLDevice if not found, no error.
out, portName, err := triggers.AutoDetectDLPIO20()
defer out.Close()

// Manual, with a remapped output window:
d, err := triggers.NewDLPIO20("/dev/ttyUSB0",
    triggers.WithIO20OutputChannels(

```

```

        triggers.IO20_AN0, triggers.IO20_AN1, triggers.IO20_AN2, triggers.IO20_AN3,
        triggers.IO20_AN4, triggers.IO20_AN5, triggers.IO20_AN6, triggers.IO20_AN7),
        triggers.WithIO20PollInterval(5*time.Millisecond),
    )
defer d.Close()

d.Send(0b00000101)           // lines 0 and 2 → AN0, AN2
d.Pulse(0, 5*time.Millisecond)
mask, err := d.ReadAll()      // bitmask of the 8 input-window channels

// Any of the 20 channels, in or out of the windows:
d.SetChannelHigh(triggers.IO20_RA4)
v, err := d.ReadChannel(triggers.IO20_AN12)

```

Channel constants are IO20\_AN0–IO20\_AN13, IO20\_RA4, IO20\_P5–IO20\_P7 (relay drivers — not TTL, not readable), IO20\_RB6, IO20\_RB7.

**Timing:** the device has no write-all command, so Send issues one 5-byte packet per line — the 8 lines change over ~3.5 ms rather than simultaneously. Use SetHigh on a single line for trigger onsets. On Linux, lower the FTDI latency timer (16 ms default) before timing-sensitive reads: `echo 1 | sudo tee /sys/bus/usb-serial/devices/ttyUSB0/latency_timer`.

### MEGTTLBox (NeuroSpin Arduino Mega TTL box)

Implements both OutputTTLDevice and InputTTLDevice. Provides 8 TTL output lines (D30–D37) and 8 TTL input lines for a FORP response pad (D22–D29).

```

box, err := triggers.NewMEGTTLBox("/dev/ttyACM0",
    triggers.WithResetDelay(2*time.Second), // wait for Arduino boot (default 2 s)
    triggers.WithPollInterval(5*time.Millisecond),
)
defer box.Close()

// Output
box.Pulse(0, 5*time.Millisecond) // pulse line 0
box.PulseMask(0b00000011, 5*time.Millisecond) // pulse lines 0 and 1
box.Send(0b00000001) // set line 0 HIGH, all others LOW

// Input (FORP response pad)
_ = box.DrainInputs(ctx) // clear latched presses from previous trial
mask, rt, err := box.WaitForInput(ctx)
buttons := triggers.DecodeMask(mask) // []FORPButton

```

**FORPButton constants** (also serve as 0-indexed line numbers for bitmask operations):

```

triggers.FORPLeftBlue   // 0
triggers.FORPLeftYellow // 1
triggers.FORPLeftGreen  // 2

```

```
triggers.FORPLeftRed    // 3
triggers.FORPRightBlue  // 4
triggers.FORPRightYellow // 5
triggers.FORPRightGreen // 6
triggers.FORPRightRed   // 7
```

## FT232HTrigger (Adafruit FT232H, Linux)

Implements both OutputTTLDevice and InputTTLDevice. Pure-Go driver via Linux usbfs — no libftdi or D2XX installation required.

**Wiring:** AD0-AD7 (D-bus) → 8 TTL output lines; AC0-AC7 (C-bus) → 8 TTL input lines.

```
box, err := triggers.NewFT232H(
    triggers.WithFT232HPollInterval(5*time.Millisecond), // optional; default 5 ms
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001) // AD0 HIGH (persistent)
box.Pulse(0, 5*time.Millisecond) // AD0: HIGH for 5 ms, then LOW
box.AllLow()

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)

// Auto-detect (falls back to NullOutputTTLDevice if no device found):
out, err := triggers.AutoDetectFT232H()
defer out.Close()
```

**Prerequisites (Linux):** - The ftdi\_sio kernel module must not hold the device: `sudo rmmod ftdi_sio` - User needs rw access to `/dev/bus/usb/BBB/DDD`. Recommended udev rule: `ACTION=="add", SUBSYSTEM=="usb", ATTRS{idVendor}=="0403", ATTRS{idProduct}=="6014", MODE="0666", GROUP="plugdev"`

## LinuxGPIOTrigger (Raspberry Pi and other Linux SBCs)

Implements both OutputTTLDevice and InputTTLDevice. Uses the Linux GPIO character device v2 API (kernel  $\geq 5.10$ ) — no external libraries required.

**Wiring:** any 8 GPIO pins for output, any other 8 for input. Pin numbers are chip-relative offsets (= BCM numbers on Raspberry Pi).

```
box, err := triggers.NewLinuxGPIOTrigger(
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),
    triggers.WithGPIOInputPins([8]int{12, 16, 20, 21, 4, 25, 24, 23}),
    triggers.WithGPIOChip("/dev/gpiochip0"), // optional; this is the default
    triggers.WithGPIOPollInterval(5*time.Millisecond), // optional; default 5 ms
)
```

```

if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001)           // pin 17 HIGH
box.Pulse(0, 5*time.Millisecond) // pin 17: HIGH for 5 ms, then LOW

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)

```

Output-only and input-only configurations are both valid — omit `WithGPIOOutputPins` or `WithGPIOInputPins` as needed.

**Prerequisites:** - Linux kernel  $\geq 5.10$  - User in the `gpio` group: `sudo usermod -aG gpio $USER`

### LabJackT4 (Modbus TCP)

Implements both `OutputTTLDevice` and `InputTTLDevice`. Pure-Go Modbus TCP driver — no LabJack SDK or system library required.

**Wiring:** the T4's digital lines are DIO4-DIO19. Output lines 0-7 → DIO4-DIO11 (FIO4-FIO7 screw terminals + EIO0-EIO3 on the DB15); input lines 0-7 → DIO12-DIO19 (EIO4-EIO7 + CIO0-CIO3). DIO0-DIO3 are the T4's dedicated analog inputs AIN0-AIN3 and cannot be used as digital lines. DIO4-DIO11 are *flexible I/O* that power up as analog inputs; NewLabJackT4 switches them to digital mode (`DIO_ANALOG_ENABLE`) before setting directions.

```

box, err := triggers.NewLabJackT4("192.168.1.100",
    triggers.WithT4PollInterval(5*time.Millisecond), // optional; default 5 ms
    triggers.WithT4Timeout(1*time.Second),           // optional; default 1 s
    triggers.WithT4UnitID(1),                         // optional; default 1
    triggers.WithT4OutputBase(4),                     // optional; DIO of output line 0
    triggers.WithT4InputBase(12),                     // optional; DIO of input line 0
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001)           // output line 0 (FIO4) HIGH
box.Pulse(0, 5*time.Millisecond) // FIO4: HIGH for 5 ms, then LOW

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)

```

The host string may include the port number ("192.168.1.100:502"); default Modbus port 502 is appended automatically if omitted.

### ParallelPort (Linux LPT)

Implements `OutputTTLDevice`.

```

ports := triggers.AvailableParallelPorts() // scans /dev/parport0..3
pp := triggers.NewParallelPort("/dev/parport0")
if err := pp.Open(); err != nil { log.Fatal(err) }
defer pp.Close()
pp.Send(0b00000111) // lines 0,1,2 HIGH
pp.Pulse(0, 10*time.Millisecond)
status, _ := pp.ReadStatus() // Linux only: status register

```

**Prerequisites:** `sudo modprobe ppdev`; user in the `lp` group.

## Null devices

`NullOutputTTLDevice` and `NullInputTTLDevice` are silent no-ops, safe to call without hardware. `AutoDetectDLPI08` and `AutoDetectFT232H` return a `NullOutputTTLDevice` when no device is found.

---

## Package `assets_embed`

Import: `github.com/chrplr/goxpyriment/assets_embed`

Provides embedded default assets as []byte slices, ready for `FontFromMemory` or `PlaySoundFromMemory`:

```

assets_embed.InconsolataFont []byte // default monospace TTF font
assets_embed.BuzzerWav       []byte // error/incorrect feedback sound
assets_embed.CorrectWav      []byte // correct/reward feedback sound

```

---

## Coordinate System

All stimulus positions are in **screen-center coordinates**:

- (0, 0) = screen center
- Positive X = right; positive Y = **up** (not down)
- `screen.CenterToSDL(x, y)` converts to SDL's top-left origin

```

      +Y (up)
      |
-X ---+--- +X
      |
      -Y (down)

```

---

## Typical Experiment Structure



```

package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("My Experiment", control.Black, control.White)
    defer exp.End()

    exp.AddDataVariableNames([]string{"block", "trial", "condition", "key", "rt_ms"})

    // Build design
    block := design.NewBlock("main")
    for _, cond := range []string{"left", "right"} {
        t := design.NewTrial()
        t.SetFactor("condition", cond)
        block.AddTrial(t, 10, false)
    }
    block.ShuffleTrials()
    exp.AddBlock(block, 1)

    err := exp.Run(func() error {
        exp.ShowInstructions("Press F for left, J for right.\n\nPress SPACE to start.")

        for bi, blk := range exp.Design.Blocks {
            for ti, trial := range blk.Trials {
                cond := trial.GetFactor("condition").(string)

                exp.Show(stimuli.NewFixCross(20, 2, control.White))
                exp.Wait(500)

                stim := stimuli.NewTextLine(cond, 0, 0, control.White)
                exp.Show(stim)

                key, rt, err := exp.Keyboard.WaitKeysRT(
                    []control.Keycode{control.K_F, control.K_J}, 3000,
                )
                if control.IsEndLoop(err) {
                    return control.EndLoop
                }

                exp.Data.Add(bi, ti, cond, key, rt)
                exp.Blank(500)
            }
        }
    })
}

```

```
    }  
    }  
    return control.EndLoop  
})  
if err != nil && !control.IsEndLoop(err) {  
    log.Fatalf("experiment error: %v", err)  
}  
}
```